2.3 Solving Equations and Transposition of Formulae

In engineering and science we often need to determine the value of an unknown quantity. For example,

$$2x + 5 = 13$$

contains the unknown value x. In many cases we must also rearrange formulae so that a different variable becomes the subject. Both skills are based on exactly the same principle.

2.3.1 The Golden Rule

Every method used to solve equations and rearrange formulae originates from one simple idea.

Definition

If two quantities are equal, then performing the same operation on both sides preserves the equality.

In other words, if

$$a = b$$

then

$$a + c = b + c$$

$$a - c = b - c$$

$$ac = bc$$

and

$$\frac{a}{c} = \frac{b}{c}$$

provided $c \neq 0$.

A common misconception is that terms are simply moved across the equals sign. In reality, the same operation is performed on both sides of the equation.

2.3.2 Solving Simple Equations

The objective when solving an equation is to leave the unknown quantity by itself.

Worked Example

Solve $2x - 4 = 10$.

Worked Example

Solve $3x + 4 = 31$.

2.3.3 Unknowns on Both Sides

Many equations contain unknown quantities on both sides. The first step is usually to gather all of the unknown terms together.

Worked Example

Solve $5x - 6 = 3x - 8$.

2.3.4 Equations Containing Brackets

When solving equations containing brackets, the brackets should usually be expanded before attempting to collect terms. Once the brackets have been expanded, the equation can be solved using the same techniques introduced earlier.

Definition

When solving equations containing brackets:

1. Expand all brackets.
2. Collect like terms.
3. Gather the unknown terms on one side.
4. Gather the constants on the other side.
5. Divide by the coefficient of the unknown.

Worked Example

Solve $3(x+4) = 21$.

First expand the bracket:

Subtract 12 from both sides:

Divide both sides by 3:

Worked Example

Solve $2(x+4) = x + 10$.

Expand the bracket:

Subtract x from both sides:

Subtract 8 from both sides:

Some equations contain brackets on both sides.

Worked Example

Solve $3(x-4) = 2(2x+1)$.

Expand both brackets:

Subtract $4x$ from both sides:

Add 12 to both sides:

Divide both sides by -1:

2.3.5 Brackets Inside Brackets

Occasionally an equation contains more than one layer of brackets. Always expand the inner brackets first.

Worked Example

Solve $3\Big(x + 2(x-1)\Big) = 18$.

Expand the inner bracket:

Simplify inside the outer bracket:

Expand:

Add 6 to both sides:

Divide by 9:

2.3.6 Checking Solutions

A useful way to check a solution is to substitute the calculated value back into the original equation.

Worked Example

The equation $2(x+4) = x + 10$ was solved, in an earlier example to give $x = 2$.

Substituting $x = 2$ into the left and right-hand sides gives

Both sides are equal, so the solution is correct.

2.3.7 Common Mistakes

Common errors when solving equations containing brackets include

- Forgetting to multiply every term inside a bracket.
- Incorrectly handling negative brackets.
- Expanding brackets before simplifying expressions inside them (while this is not an error in itself, not simplifying can lead to errors).
- Collecting unlike terms.
- Solving only part of the equation.
- Forgetting to check the final answer.

2.3.8 What is Transposition?

So far we have used algebra to solve equations containing a single unknown. However, engineers often work with formulae that contain several variables. For example, $F = ma$, contains the variables F, m and a. Sometimes the required variable is already isolated. For example, the formula above already gives force F in terms of mass m and acceleration a. In other situations, a different variable may be required. For example, if the force and acceleration are known, we may wish to find the mass instead. This requires the formula to be rearranged.

Definition

Transposition is the process of rearranging a formula so that a different variable becomes the subject. All transpositions are performed using the Golden Rule:

Whatever operation is performed on one side of an equation must also be performed on the other side.

When transposing a formula, the objective is to leave the required variable by itself on one side of the equation.

Worked Example

Consider the formula $F = ma$. If we wish to make m the subject, begin by dividing both sides by a

Simplifying gives

It is usually written as

Notice that the variable has not been "moved across" the equals sign. Instead, the same operation was performed on both sides of the equation.

Many transpositions follow a similar process:

- Remove terms using addition or subtraction.
- Remove factors using multiplication or division.
- Remove powers using roots.
- Remove roots using powers.

The order in which these operations are reversed is often the reverse of the order in which they were originally applied.

Transposition is an essential engineering skill because many formulae can be used to calculate several different quantities. Rather than memorising a separate formula for every situation, engineers often rearrange existing formulae to obtain the required variable.

2.3.9 Rearranging Simple Formulae

When solving an equation, we usually wish to find the value of an unknown quantity. When rearranging a formula, the objective is slightly different. Instead of finding a numerical value, we wish to make a particular variable the subject of the formula. The same algebraic principles are used in both cases. The key idea is to remove anything attached to the required variable until it is left by itself.

Definition

To make a variable the subject of a formula:

1. Identify the variable that is to become the subject.
2. Remove terms using addition or subtraction.
3. Remove factors using multiplication or division.
4. Continue until the variable appears by itself.

Worked Example

Make a the subject of $\mathrm{v} = \mathrm{u} + at$.

Therefore,

Worked Example

Make R the subject of $V = IR$.

Therefore,

This relationship is frequently encountered in electrical engineering as a rearrangement of Ohm's Law.

Worked Example

Make h the subject of $A = 2\pi r^2 + 2\pi r h$.

Therefore,

Worked Example

Make x the subject of $y = 3x + 7$.

Therefore,

Worked Example

Make t the subject of $s = ut$.

Therefore,

Notice that every example follows the same pattern:

1. Remove additions and subtractions first.
2. Remove multiplications and divisions second.
3. Continue until the required variable is isolated.

More complicated formulae may contain fractions, brackets, powers or roots, but the same fundamental principle always applies:

Perform the same operation on both sides of the equation.

2.3.10 Formulae Containing Fractions

Many engineering formulae contain fractions. Although these formulae may appear more complicated, they are rearranged using exactly the same principles as before. The main difficulty is that the required variable often appears inside a fraction. In many cases, the simplest approach is to remove fractions as early as possible by multiplying both sides of the equation.

Definition

When rearranging formulae containing fractions:

1. Remove fractions whenever possible.
2. Multiply both sides by the denominator.
3. Expand any brackets that remain.
4. Collect terms containing the required variable.
5. Isolate the required variable.

Worked Example

Make t the subject of $\frac{2+t}{3} = 2(t-k)$.

Therefore,

Worked Example

Make x the subject of $\frac{x+4}{3} = 7$.

Sometimes the variable appears in the denominator of a fraction. The same algebraic rules apply, but multiplication is often used to remove the denominator.

Worked Example

Make μ the subject of $\frac{1}{f} = \frac{1}{u} + \frac{1}{\mu}$.

This is a valid answer, but it can be simplified further:

Worked Example

Make R the subject of $E = V\left(\frac{R+r}{R}\right)$.

2.3.10.1 Key Observation

Fractions often look intimidating because they contain several layers of operations. However, the same algebraic ideas are still being used:

- Multiply to remove denominators.
- Expand brackets when necessary.
- Gather like terms.

- Factorise where appropriate.
- Isolate the required variable.

Once fractions have been removed, most rearrangements become ordinary algebraic problems.

2.3.11 Formulae Containing Roots and Powers

Some engineering formulae contain powers, indices or roots. These formulae are rearranged using the same principles as before. However, additional operations may be required to remove roots or powers. The key idea is to reverse the operation that is applied to the required variable. For example:

- A square root can be removed by squaring.
- A cube root can be removed by cubing.
- A square can be removed by taking a square root.
- Higher powers can be removed using the appropriate root.

Definition

Roots and powers are inverse operations.

For example,

$$(\sqrt{x})^2 = x$$

and

$$\sqrt{x^2} = x$$

When rearranging formulae, powers and roots can therefore be removed by applying the inverse operation to both sides of the equation.

Worked Example

Make x the subject of $y = x^3$.

Therefore,

Worked Example

Make r the subject of $A = \pi r^2$.

Therefore,

Worked Example

Make r the subject of $V = \frac{4}{3}\pi r^3$.

Therefore,

Worked Example

Make L the subject of $T = 2\pi\sqrt{\frac{L}{g}}$.

Therefore,

Worked Example

Make x the subject of $y = 5 + \sqrt{x-2}$.

Therefore,

2.3.11.1 Common Mistakes with Roots and Powers

Students often make mistakes when removing roots and powers. Common errors include:

- Forgetting to apply the operation to both sides.
- Taking a square root when a cube root is required.
- Forgetting that square roots and squares are inverse operations.
- Squaring only part of an expression.
- Omitting brackets when squaring expressions.

Worked Example

Consider $y - 5 = \sqrt{x-2}$.

A common mistake is to write

$$y^2 - 5^2 = x - 2$$

This is incorrect. The entire expression must be squared:

$$(y-5)^2 = x - 2$$

2.3.12 Common Mistakes

Rearranging formulae is largely a matter of applying the Golden Rule correctly. Many mistakes occur not because the mathematics is difficult, but because a small algebraic error is made during the rearrangement process. Whenever a formula has been rearranged, it is good practice to check that

- the same operation has been performed on both sides,
- brackets have been handled correctly,
- negative signs have not been lost,
- fractions have been treated correctly,
- the final answer can be substituted back into the original formula.

The following examples illustrate some common mistakes.

Mistake 1: Performing Different Operations on Each Side

Worked Example

Consider $2x + 5 = 17$. A common mistake is

$$2x = 17$$

This is incorrect because 5 has only been removed from the left-hand side.

The correct method is

$$2x + 5 - 5 = 17 - 5$$

Therefore,

$$2x = 12$$

Mistake 2: Forgetting to Expand All Terms in a Bracket

Worked Example

Consider $3(x+4)$. A common mistake is

$$3x + 4$$

This is incorrect because the 3 must multiply every term inside the bracket.

The correct expansion is

$$3x + 12$$

Mistake 3: Losing Negative Signs

Negative signs are one of the most common sources of errors.

Worked Example

Consider $5x - (2x + 3)$. A common mistake is

$$5x - 2x + 3$$

This is incorrect. The negative sign applies to every term inside the bracket. The correct result is

$$5x - 2x - 3$$

Always think of the negative sign as multiplying the entire bracket by -1.

Mistake 4: Cancelling Terms Instead of Factors

Cancelling is only valid when quantities appear as factors.

Worked Example

Consider $\frac{3x}{3}$. Since 3 is a factor in both the numerator and denominator,

$$\frac{3x}{3} = x$$

This is correct.

However,

Worked Example

Consider $\frac{x+3}{3}$. A common mistake is to cancel the 3's and write

$$x + 1$$

This is incorrect. The numerator contains two separate terms, not a factor of 3. The expression cannot be simplified in this way.

Mistake 5: Multiplying Only Part of a Fractional Equation

Worked Example

Consider $\frac{x+2}{3} = 4$. To remove the denominator, we multiply both sides by 3:

$$3\left(\frac{x+2}{3}\right) = 3(4)$$

Therefore,

$$x + 2 = 12$$

A common mistake is to multiply only the numerator by 3, which does not preserve the equality. Always multiply the entire side of the equation.

Mistake 6: Incorrectly Removing Roots

Worked Example

Consider $y = \sqrt{x}$. A common mistake is to subtract the square root symbol and write

$$y = x$$

This is incorrect. The square root is removed by squaring both sides:

$$y^2 = (\sqrt{x})^2$$

Therefore,

$$y^2 = x$$

Mistake 7: Forgetting Brackets When Squaring

Worked Example

Consider $y - 5 = \sqrt{x}$. A common mistake is

$$y^2 - 25 = x$$

This is incorrect because only part of the expression has been squared. The entire left-hand side must be squared:

$$(y - 5)^2 = x$$

If an expression contains more than one term, use brackets.

Final Check

After solving an equation or rearranging a formula, ask the following questions:

1. Have I performed the same operations on both sides?
2. Have I expanded brackets correctly?
3. Have I kept all negative signs?
4. Have I handled fractions correctly?
5. Is the required variable completely isolated?
6. Can I substitute the result back into the original equation to verify it?

Most algebraic mistakes can be detected by performing this simple final check.

2.3.13 Engineering Application

Engineering Application

One of the most important uses of algebra in engineering is the rearrangement of formulae.

Engineers rarely memorise a separate equation for every possible calculation. Instead, a single formula is rearranged to make the required variable the subject.

This approach reduces the number of formulae that must be remembered and allows the same physical relationship to be used in different situations.

Consider Newton's Second Law: $F = ma$. The formula can be used in three different ways:

$$F = ma,$$

$$m = \frac{F}{a},$$

and

$$a = \frac{F}{m}$$

The same equation can therefore be used to calculate force, mass or acceleration.

In many practical situations the required quantity is not the original subject of the formula. For example:

$$V = IR$$

may be rearranged to find R and

$$T = 2\pi\sqrt{\frac{L}{g}}$$

may be rearranged to find L.

The ability to rearrange and manipulate formulae is therefore one of the most important algebraic skills in engineering mathematics. Rather than memorising many different versions of the same equation, engineers use algebra to obtain the version they need.

2.3.14 Exercises

Exercise

1. Solve the following equations:

   (a) $6x - 4 = 8$

   (b) $8q + 6 = 4q - 14$

   (c) $14 + 13y = 20y - 21$

   (d) $-15b + 21 + 5b = -19$

   (e) $-10x + 90 = -21x + 2$

2. Solve the following equations containing brackets:

   (a) $3(x + 10) = 63$

   (b) $-3(7p + 5) = 27$

   (c) $7 - (5t - 13) = -25$

   (d) $2(h + 4) = 3(h + 10) - 2$

   (e) $3\left(x + 2(x - 2) - 2\left(x - 3(x - 1)\right)\right) = 0$

   (f) $4(x^2 + 2) = 44$

3. Solve the following equations:

   (a) $\dfrac{-8 - 3k}{2} = 11$

   (b) $9 = \dfrac{p + 4}{p + 12}$

   (c) $\dfrac{5b + 10}{5} = -b + 10$

   (d) $\dfrac{3y + 2}{2} = 6y + 4$

   (e) $\dfrac{7x}{4} - 3 = 2 + \dfrac{9x}{2}$

4. Make u the subject of $s = ut + \frac{1}{2}at^2$

5. Make x the subject of $m = k\sqrt{a(1 - x)}$

6. Make l the subject of $V = \pi r^2 l + \frac{1}{3}\pi r^2 h$

7. Make c_1 the subject of $P = \mu_1 c_1 + \mu_2 c_2$

Exercise

8. Make M the subject of $\rho = \frac{M}{V}$
9. Make A the subject of $T = \frac{V}{A} + d$
10. Make x the subject of $F = \frac{x}{k} + E$
11. Make I the subject of $V = \frac{jI}{\omega C} + V_1$
12. Make r the subject of $V = \dfrac{4}{3}\pi r^3$
13. Make x the subject of $y = 5 + \sqrt{x-2}$
14. Make e the subject of:

$$T = \frac{2v}{g}\left(\frac{1}{1-e}\right)$$

15. Given that:

$$U = V\left(1 - \frac{C}{D\sqrt{N}}\right),$$

find N in terms of U, V, D and C.

16. Make b the subject of:

$$a(3b-1) = 2b + 2$$

17. The equations for a battery with e.m.f. E and internal resistance r connected across a resistor R can be expressed as $E = V\left(\frac{R+r}{R}\right)$. Determine an expression for R.
18. Make r the subject of:

$$P = \frac{P_0}{1-r^2}$$

19. Make x the subject of:

$$y = a + \frac{1}{1-x}$$

20. The period of a pendulum is given by

$$T = 2\pi\sqrt{\frac{L}{g}}$$

Given $T = 2.84$ s and $g = 9.81$ m/s^2, determine the length L.

Solutions to Exercise 2.3.14

1. (a) Move all constants to the right-hand side (RHS):

$$\begin{aligned}
6x - 4 &= 8 && \text{[add 4 to both sides]} \\
6x &= 12 && \text{[divide both sides by 6]} \\
\therefore \quad x &= \frac{12}{6} = 2
\end{aligned}$$

   (b) This solution demonstrates that it doesn't matter what side we take the variable to (try it the other way to confirm that you get the same answer):

$$\begin{aligned}
8q + 6 &= 4q - 14 && \text{[subtract } 8q \text{ from both sides]} \\
6 &= -4q - 14 && \text{[add 14 to both sides]} \\
20 &= -4q && \text{[divide both sides by -4]} \\
\therefore \quad q &= -5 && \text{[note that this is the same as } -5 = q\text{]}
\end{aligned}$$

   (c) Gather 'like' terms:

$$\begin{aligned}
14 + 13y &= 20y - 21 && \text{[subtract } 20y \text{ from both sides]} \\
14 - 7y &= -21 && \text{[subtract 14 from both sides]} \\
-7y &= -35 && \text{[divide both sides by -7]} \\
\therefore \quad y &= 5
\end{aligned}$$

   (d) Gather 'like' terms:

$$\begin{aligned}
-15b + 21 + 5b &= -19 && \text{[subtract 21 from both sides]} \\
-10b &= -40 && \text{[divide both sides by -10]} \\
\therefore \quad b &= 4
\end{aligned}$$

(e) Gather 'like' terms:

$$\begin{aligned}
-10x + 90 &= -21x + 2 && \text{[add } 21x \text{ to both sides]} \\
11x + 90 &= 2 && \text{[subtract 90 from both sides]} \\
11x &= -88 && \text{[divide both sides by 11]} \\
\therefore \quad x &= -8
\end{aligned}$$

2. (a) Multiply out the brackets first, as the variable we wish to solve for is inside the bracket:

$$\begin{aligned}
3(x + 10) &= 63 \\
3x + 30 &= 63 && \text{[subtract 30 from both sides]} \\
3x &= 33 && \text{[divide both sides by 3]} \\
\therefore \quad x &= 11
\end{aligned}$$

(b)

$$\begin{aligned}
-3(7p + 5) &= 27 && \text{[multiply out the brackets]} \\
-21p - 15 &= 27 && \text{[add 15 to both sides]} \\
-21p &= 42 && \text{[divide both sides by -21]} \\
\therefore \quad p &= -2
\end{aligned}$$

(c)

$$\begin{aligned}
7 - (5t - 13) &= -25 && \text{[multiply out the brackets]} \\
7 - 5t + 13 &= -25 && \text{[simplify]} \\
-5t + 20 &= -25 && \text{[subtract 20 from both sides]} \\
-5t &= -45 && \text{[divide both sides by -5]} \\
\therefore \quad t &= 9
\end{aligned}$$

(d)

$$\begin{aligned}
2(h+4) &= 3(h+10)-2 && \text{[multiply out the brackets]}\\
2h+8 &= 3h+30-2 && \text{[simplify]}\\
2h+8 &= 3h+28 && \text{[subtract } 3h \text{ from both sides]}\\
-h+8 &= 28 && \text{[subtract 8 from both sides]}\\
-h &= 20 && \text{[divide both sides by -1]}\\
\therefore \quad h &= -20
\end{aligned}$$

(e)

$$\begin{aligned}
3\big(x+2(x-2)\big)-2\big(x-3(x-1)\big) &= 0 && \text{[multiply out the inner brackets first]}\\
3\big(x+2x-4\big)-2\big(x-3x+3\big) &= 0 && \text{[multiply out the outer brackets next]}\\
3x+6x-12-2x+6x-6 &= 0 && \text{[simplify]}\\
13x-18 &= 0 && \text{[add 18 to both sides]}\\
13x &= 18 && \text{[divide both sides by 13]}\\
\therefore \quad x &= \frac{18}{13}
\end{aligned}$$

(f)

$$\begin{aligned}
4(x^2+2) &= 44 && \text{[divide both sides by 4]}\\
x^2+2 &= 11 && \text{[subtract 2 from both sides]}\\
x^2 &= 9 && \text{[square root both sides]}\\
\therefore \quad x &= \pm 3
\end{aligned}$$

3. (a)

$$\begin{aligned}
\frac{-8-3k}{2} &= 11 && \text{[muliply both sides by 2]}\\
-8-3k &= 22 && \text{[add 8 to both sides]}\\
-3k &= 30 && \text{[divide both sides by -3]}\\
\therefore \quad k &= -10
\end{aligned}$$

(b)

$$\begin{aligned}
9 &= \frac{p+4}{p+12} && \text{[multiply both sides by } p+12\text{]}\\
9(p+12) &= p+4 && \text{[multiply out the brackets]}\\
9p+108 &= p+4 && \text{[subtract } p \text{ from both sides]}\\
8p+108 &= 4 && \text{[subtract 108 from both sides]}\\
8p &= -104 && \text{[divide both sides by 8]}\\
\therefore \quad p &= -13
\end{aligned}$$

(c)

$$\begin{aligned}
\frac{5b+10}{5} &= -b+10 && \text{[multiply both sides by 5]}\\
5b+10 &= 5(-b+10) && \text{[multiply out the brackets]}\\
5b+10 &= -5b+50 && \text{[add } 5b \text{ to both sides]}\\
10b+10 &= 50 && \text{[subtract 10 from both sides]}\\
10b &= 40 && \text{[divide both sides by 10]}\\
\therefore \quad b &= 4
\end{aligned}$$

(d)

$$\begin{aligned}
\frac{3y+2}{2} &= 6y+4 && \text{[multiply both sides by 2]}\\
3y+2 &= 2(6y+4) && \text{[multiply out the brackets]}\\
3y+2 &= 12y+8 && \text{[subtract } 12y \text{ from both sides]}\\
-9y+2 &= 8 && \text{[subtract 2 from both sides]}\\
-9y &= 6 && \text{[divide both sides by -9]}\\
\therefore \quad y &= \frac{6}{-9} = -\frac{2}{3}
\end{aligned}$$

(e) Here the fraction doesn't take up the whole of either the LHS or RHS, so when we multiply both sides by a number, it means multiplying every single term on each side:

$$
\begin{aligned}
\frac{7x}{4} - 3 &= 2 + \frac{9x}{2} && \text{[multiply both sides by 4]} \\
4\left(\frac{7x}{4} - 3\right) &= 4\left(2 + \frac{9x}{2}\right) && \text{[multiply out the brackets]} \\
7x - 12 &= 8 + 18x && \text{[Here } 4 \times 9x \text{ is } 36x\text{, but then we divide by 2]}
\end{aligned}
$$

If the denominator still remained, we would still have a fraction on the RHS. If that is the case, then multiplying by that denominator would be necessary.

$$
\begin{aligned}
7x - 12 &= 8 + 18x && \text{[subtract } 18x \text{ from both sides]} \\
-11x - 12 &= 8 && \text{[add 12 to both sides]} \\
-11x &= 20 && \text{[divide both sides by -11]} \\
\therefore \quad x &= -\frac{20}{11}
\end{aligned}
$$

4.

$$
\begin{aligned}
s &= ut + \frac{1}{2}at^2 && \text{[subtract } \tfrac{1}{2}at^2 \text{ from both sides]} \\
ut &= s - \frac{1}{2}at^2 && \text{[divide both sides by } t\text{]} \\
u &= \frac{s - \frac{1}{2}at^2}{t} && \text{[split the numerator]} \\
u &= \frac{s}{t} - \frac{\frac{1}{2}at^2}{t} && \text{[simplify]} \\
\therefore u &= \frac{s}{t} - \frac{1}{2}at && \text{[this could also be written as]} \\
u &= \frac{s}{t} - \frac{at}{2}
\end{aligned}
$$

5. As with the majority of equations, there is more than one way to transpose this equation.

$$
\begin{aligned}
m &= k\sqrt{a(1-x)} && \text{[divide both sides by } k\text{]} \\
\frac{m}{k} &= \sqrt{a(1-x)} && \text{[square both sides]} \\
\left(\frac{m}{k}\right)^2 &= a(1-x) && \text{[we can square each element]} \\
\frac{m^2}{k^2} &= a(1-x) && \text{[divide both sides by } a\text{]} \\
\frac{\frac{m^2}{k^2}}{a} &= 1-x && \text{[which can be written as]} \\
\frac{m^2}{ak^2} &= 1-x && \text{[add } x \text{ to both sides]} \\
\frac{m^2}{ak^2} + x &= 1 && \text{[subtract } \frac{m^2}{ak^2} \text{ from both sides]} \\
\therefore \qquad x &= 1 - \frac{m^2}{ak^2}
\end{aligned}
$$

6.

$$
\begin{aligned}
V &= \pi r^2 l + \frac{1}{3}\pi r^2 h && \text{[subtract } \frac{1}{3}\pi r^2 h \text{ from both sides]} \\
\pi r^2 l &= V - \frac{1}{3}\pi r^2 h && \text{[divide both sides by } \pi r^2\text{]} \\
l &= \frac{V - \frac{1}{3}\pi r^2 h}{\pi r^2} && \text{[split the numerator]} \\
l &= \frac{V}{\pi r^2} - \frac{\frac{1}{3}\pi r^2 h}{\pi r^2} && \text{[simplify]} \\
\therefore l &= \frac{V}{\pi r^2} - \frac{h}{3}
\end{aligned}
$$

7.

$$
\begin{aligned}
P &= \mu_1 c_1 + \mu_2 c_2 && \text{[subtract } \mu_2 c_2 \text{ from both sides]} \\
P - \mu_2 c_2 &= \mu_1 c_1 && \text{[divide both sides by } \mu_1\text{]} \\
\therefore \qquad c_1 &= \frac{P - \mu_2 c_2}{\mu_1}
\end{aligned}
$$

8.

$$\begin{aligned} \rho &= \frac{M}{V} && \text{[multiply both sides by } V\text{]} \\ \therefore M &= \rho V \end{aligned}$$

9.

$$\begin{aligned} T &= \frac{V}{A} + d && \text{[subtract } d \text{ from both sides]} \\ \frac{V}{A} &= T - d && \text{[multiply both sides by } A\text{]} \\ \therefore V &= A(T-d) && \text{[divide both sides by } (T-d)\text{]} \\ \therefore A &= \frac{V}{T-d} \end{aligned}$$

Note that this could have been transposed differently. Some people wish to get rid of the fraction first: This is fine to do, but remember that *both sides* should be multiplied by A. For example:

$$\begin{aligned} T &= \frac{V}{A} + d && \text{[multiply both sides by } A\text{]} \\ AT &= A\left(\frac{V}{A} + d\right) && \text{[multiply out the brackets]} \\ AT &= V + Ad && \text{[subtract } Ad \text{ from both sides]} \\ V &= AT - Ad && \text{[factorise the right-hand side]} \\ \therefore \quad V &= A(T-d) && \text{[divide both sides by } (T-d)\text{]} \\ \therefore A &= \frac{V}{T-d} \end{aligned}$$

10.

$$\begin{aligned} F &= \frac{x}{k} + E && \text{[subtract } E \text{ from both sides]} \\ F - E &= \frac{x}{k} && \text{[multiply both sides by } k\text{]} \\ \therefore \quad x &= k(F-E) \end{aligned}$$

11.

$$
\begin{aligned}
V &= \frac{jI}{\omega C} + V_1 && \text{[subtract } V_1 \text{ from both sides]} \\
\frac{jI}{\omega C} &= V - V_1 && \text{[multiply both sides by } \omega C\text{]} \\
jI &= \omega C\,(V - V_1) && \text{[divide both sides by } j\text{]} \\
\therefore \quad I &= \frac{\omega C\,(V - V_1)}{j}
\end{aligned}
$$

12.

$$
\begin{aligned}
V &= \frac{4}{3}\pi r^3 && \text{[multiply both sides by 3]} \\
3V &= 4\pi r^3 && \text{[divide both sides by } 4\pi\text{]} \\
r^3 &= \frac{3V}{4\pi} && \text{[cube root both sides]} \\
\therefore \quad r &= \sqrt[3]{\frac{3V}{4\pi}}
\end{aligned}
$$

13.

$$
\begin{aligned}
y &= 5 + \sqrt{x-2} && \text{[subtract 5 from both sides]} \\
y - 5 &= \sqrt{x-2} && \text{[square both sides]} \\
(y-5)^2 &= x - 2 && \text{[add 2 to both sides]} \\
\therefore \quad x &= (y-5)^2 + 2
\end{aligned}
$$

14.

$$
\begin{aligned}
T &= \frac{2v}{g}\left(\frac{1}{1-e}\right) && \text{[multiply both sides by } 1-e\text{]}\\
(1-e)\,T &= \frac{2v}{g} && \text{[divide both sides by } T\text{]}\\
1-e &= \frac{2v}{gT} && \text{[add } e \text{ to both sides]}\\
1 &= \frac{2v}{gT}+e && \text{[subtract } \tfrac{2v}{gT} \text{ from both sides]}\\
\therefore \quad e &= 1-\frac{2v}{gT}
\end{aligned}
$$

15. We will “unwrap” N, first by dividing both sides by V:

$$\therefore\ \frac{U}{V} = 1 - \frac{C}{D\sqrt{N}}$$

Isolate the term containing N:

$$\therefore\ \frac{U}{V} - 1 = -\frac{C}{D\sqrt{N}}$$

Remove N from the denominator of the fraction by multiplication:

$$\therefore\ \sqrt{N}\left(\frac{U}{V} - 1\right) = -\frac{C}{D}$$

Isolate the $\sqrt{N}$ and simplify the RHS:

$$\therefore\ \sqrt{N} = \frac{-\dfrac{C}{D}}{\dfrac{U}{V} - 1}$$

$$= \frac{-C}{D\left(\dfrac{U}{V} - 1\right)}$$

$$= \frac{-C}{-D\left(1 - \dfrac{U}{V}\right)}$$

$$= \frac{C}{D\left(1 - \dfrac{U}{V}\right)}$$

$$= \frac{CV}{D(V - U)}$$

Finally square both sides to obtain N:

$$\therefore\ N = \left(\frac{CV}{D(V - U)}\right)^2$$

16.

$$\begin{aligned}
a(3b - 1) &= 2b + 2 && \text{[multiply out the brackets]}\\
3ab - a &= 2b + 2 && \text{[subtract } 2b \text{ from both sides]}\\
3ab - 2b - a &= 2 && \text{[add } a \text{ to both sides]}\\
3ab - 2b &= 2 + a && \text{[factorise the LHS]}\\
b(3a - 2) &= 2 + a && \text{[divide both sides by } 3a - 2\text{]}\\
\therefore \quad b &= \frac{2 + a}{3a - 2}
\end{aligned}$$

17.

$$
\begin{aligned}
E &= \frac{V(R+r)}{R} && \text{[multiply both sides by } R\text{]} \\
ER &= V(R+r) && \text{[multiply out the brackets]} \\
ER &= VR + Vr && \text{[subtract } VR \text{ from both sides]} \\
ER - VR &= Vr && \text{[factorise the LHS]} \\
R(E-V) &= Vr && \text{[divide both sides by } E-V\text{]} \\
\therefore \quad R &= \frac{Vr}{E-V}
\end{aligned}
$$

18.

$$
\begin{aligned}
P &= \frac{P_0}{1-r^2} && \text{[multiply both sides by } 1-r^2\text{]} \\
P(1-r^2) &= P_0 && \text{[multiply out the brackets]} \\
P - Pr^2 &= P_0 && \text{[subtract } P \text{ from both sides]} \\
-Pr^2 &= P_0 - P && \text{[divide both sides by } -P\text{]} \\
r^2 &= \frac{P_0 - P}{-P} && \text{[simplify]} \\
r^2 &= \frac{P - P_0}{P} && \text{[square root both sides]} \\
\therefore \quad r &= \pm\sqrt{\frac{P-P_0}{P}}
\end{aligned}
$$

19.

$$
\begin{aligned}
y &= a + \frac{1}{1-x} && \text{[subtract } a \text{ from both sides]} \\
y - a &= \frac{1}{1-x} && \text{[multiply both sides by } 1-x\text{]} \\
(1-x)(y-a) &= 1 && \text{[divide both sides by } y-a\text{]} \\
1 - x &= \frac{1}{y-a} && \text{[subtract 1 from both sides]} \\
-x &= \frac{1}{y-a} - 1 && \text{[divide both sides by } -1\text{]} \\
\therefore \quad x &= 1 - \frac{1}{y-a}
\end{aligned}
$$

20. There are two ways to approach this: either rearrange for L first and then substitute the values or the opposite way around. I'm going with the latter approach:

$$
\begin{aligned}
T &= 2\pi\sqrt{\frac{L}{g}} && \text{[substitute the values]} \\
2.84 &= 2\pi\sqrt{\frac{L}{9.81}} && \text{[divide by } 2\pi\text{]} \\
0.452 &= \sqrt{\frac{L}{9.81}} && \text{[square both sides]} \\
0.204 &= \frac{L}{9.81} && \text{[multiply by 9.81]} \\
L &= 2.00 \text{ m to 2 decimal places}
\end{aligned}
$$